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- (54) **AREA-BASED IDENTIFICATION OF DEVICES FOR PAIRING**
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CPC **G06Q 20/201** (2013.01); **G06Q 20/208** (2013.01)
(58) **Field of Classification Search**
None
See application file for complete search history.
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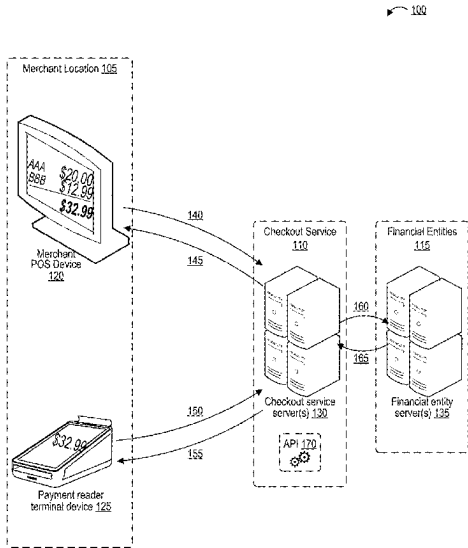
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(57) **ABSTRACT**
A merchant uses a merchant point of sale (POS) device to identify items or services for purchase by a customer, and costs for each. One or more payment amounts are sent from the merchant POS device to a checkout server using an application programming interface (API), which the checkout server sends on to an authorized payment terminal device that then reads payment information from a payment object and sends the payment information back to the checkout server. The checkout server processes the transaction and confirms transaction completion to the merchant POS device and payment terminal device.
- 20 Claims, 7 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/533,095, filed on
Aug. 6, 2019, now Pat. No. 11,328,277.

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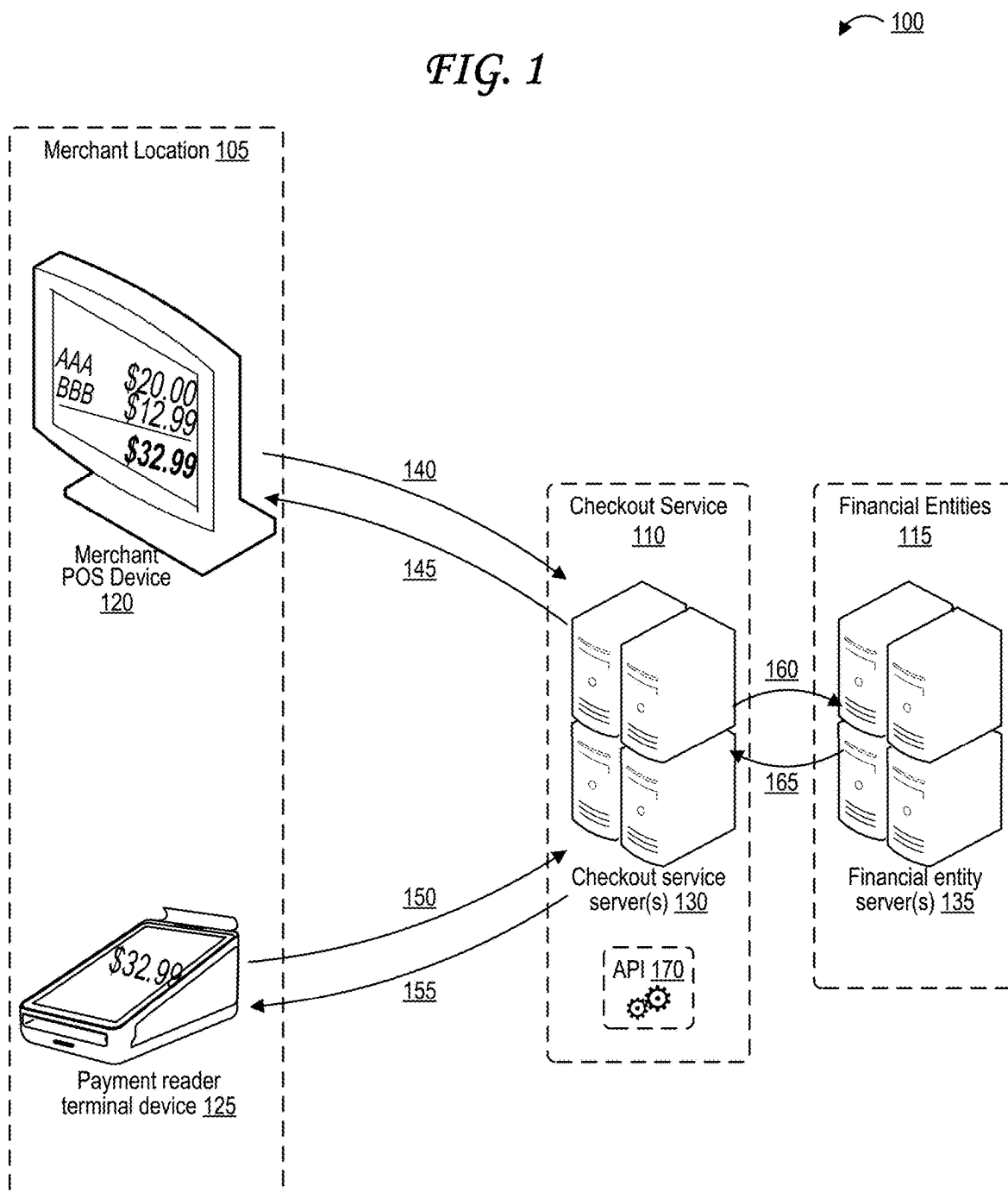
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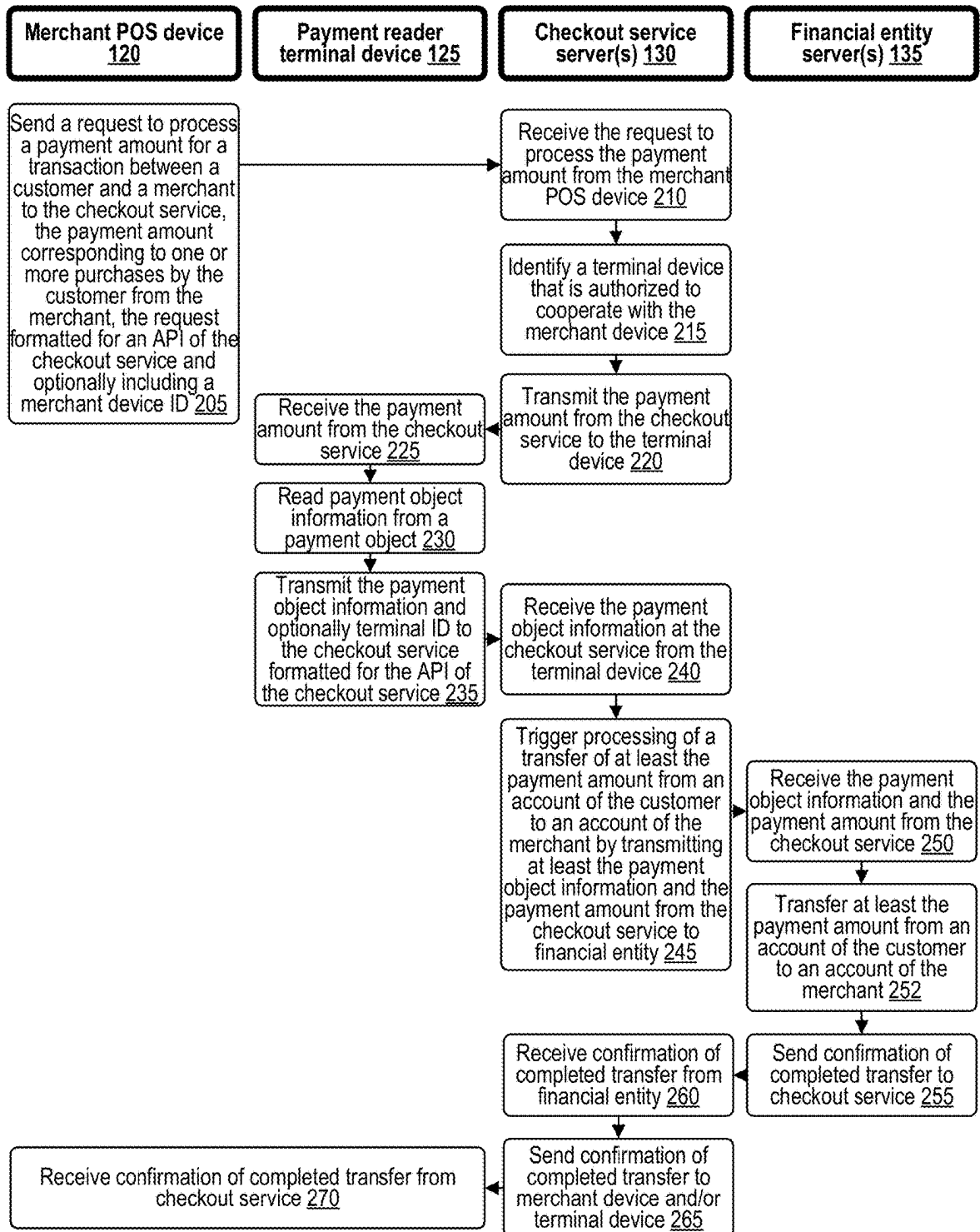
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FIG. 1



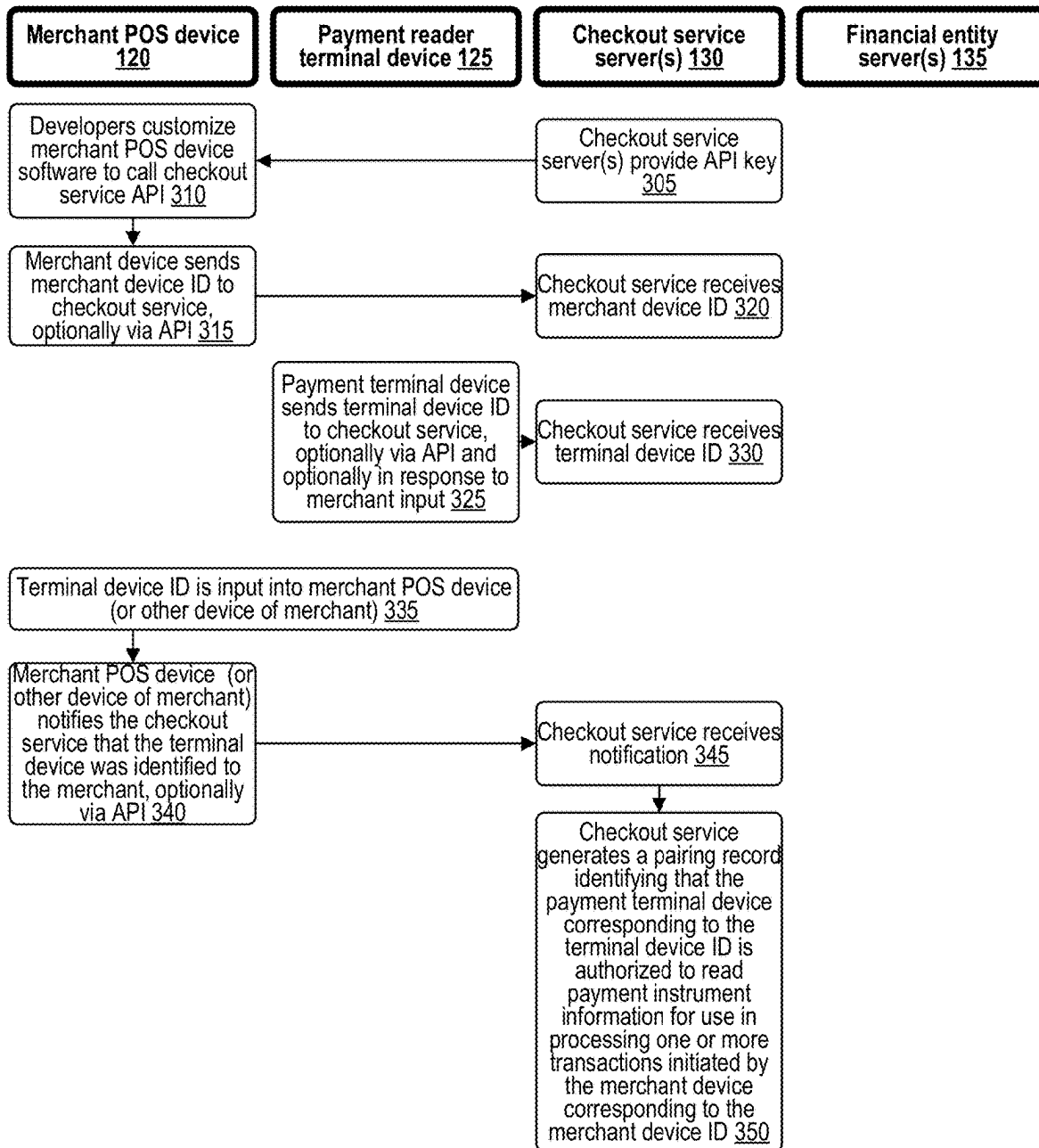
↪ 200

FIG. 2



300

FIG. 3



400

FIG. 4

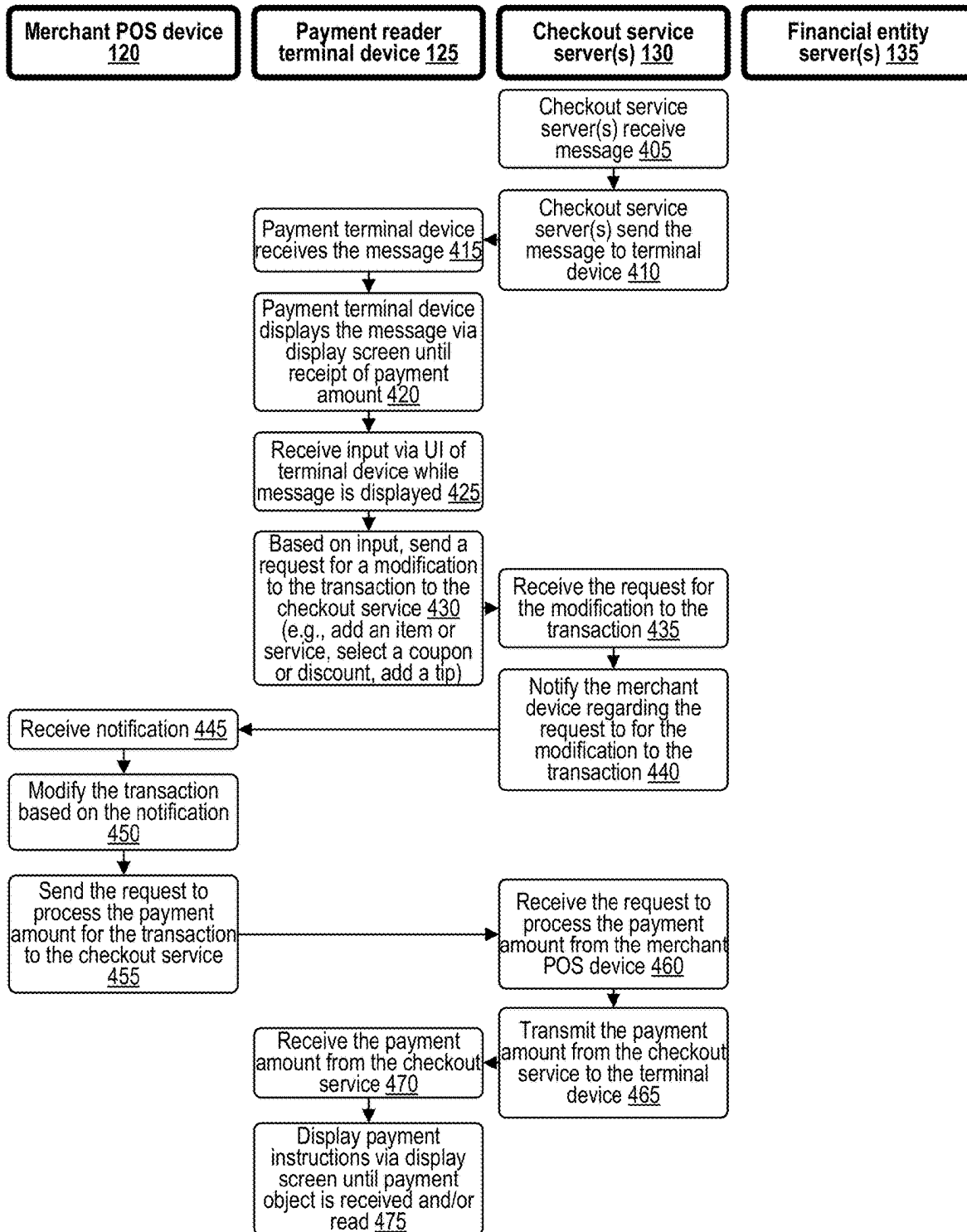
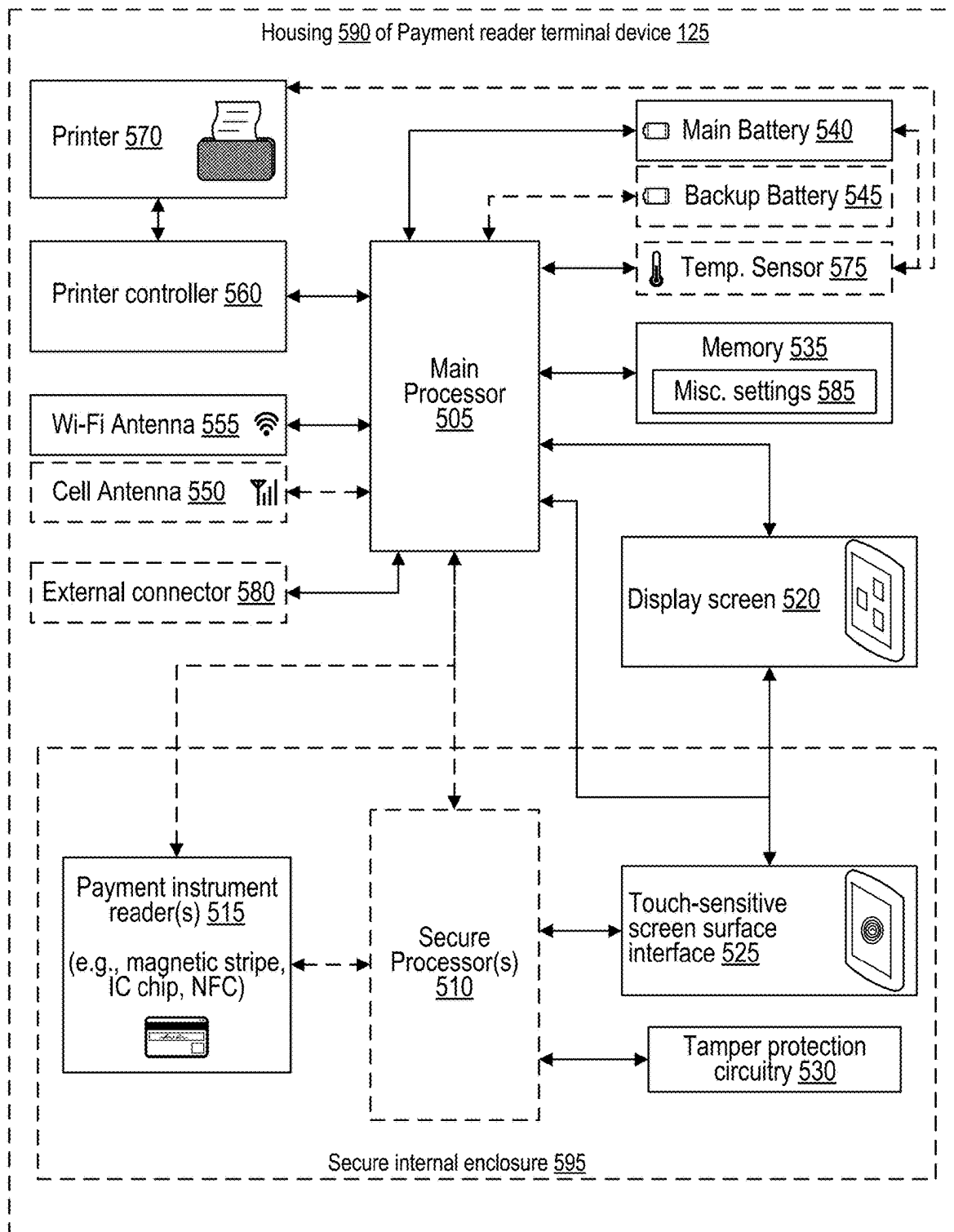


FIG. 5

500



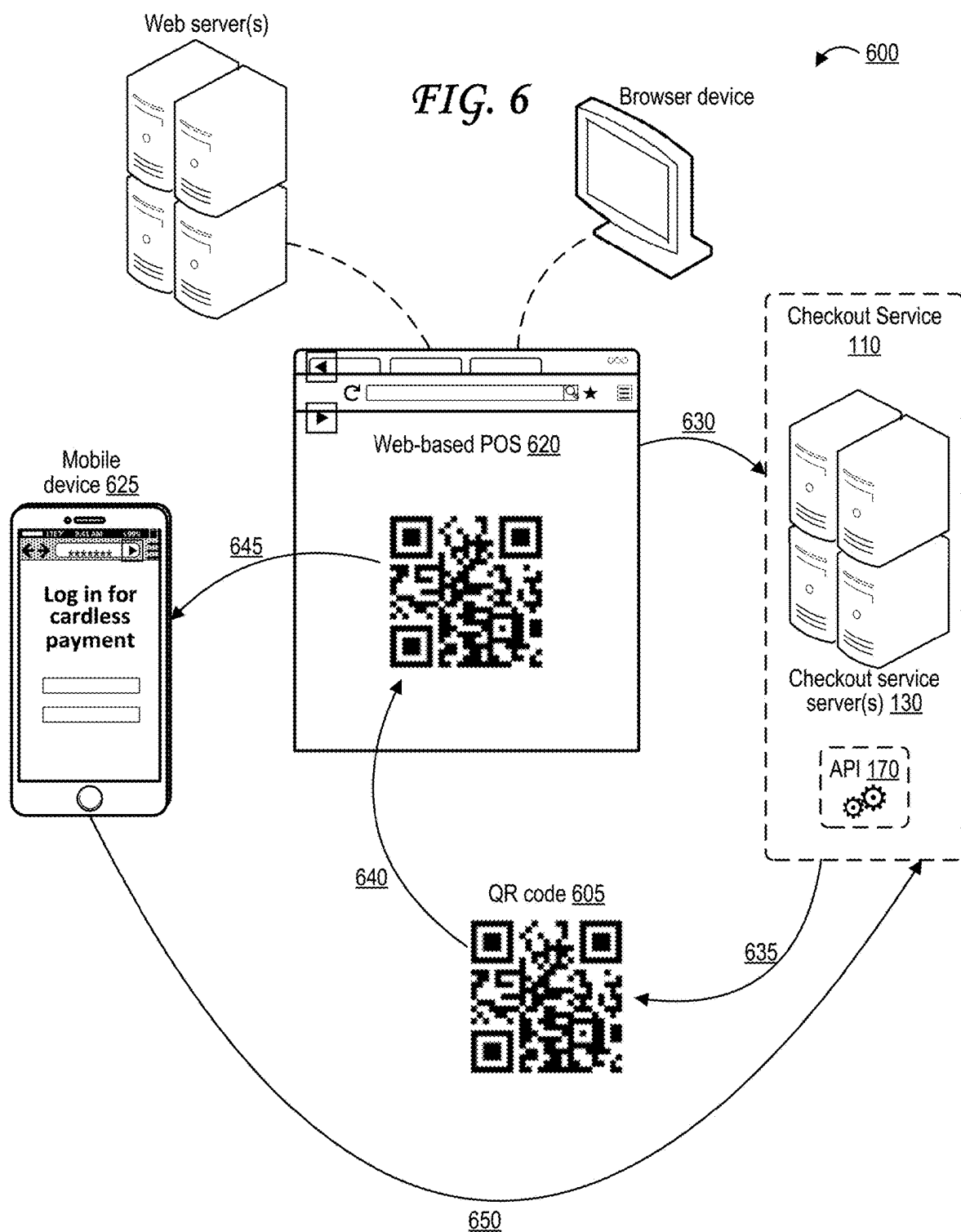
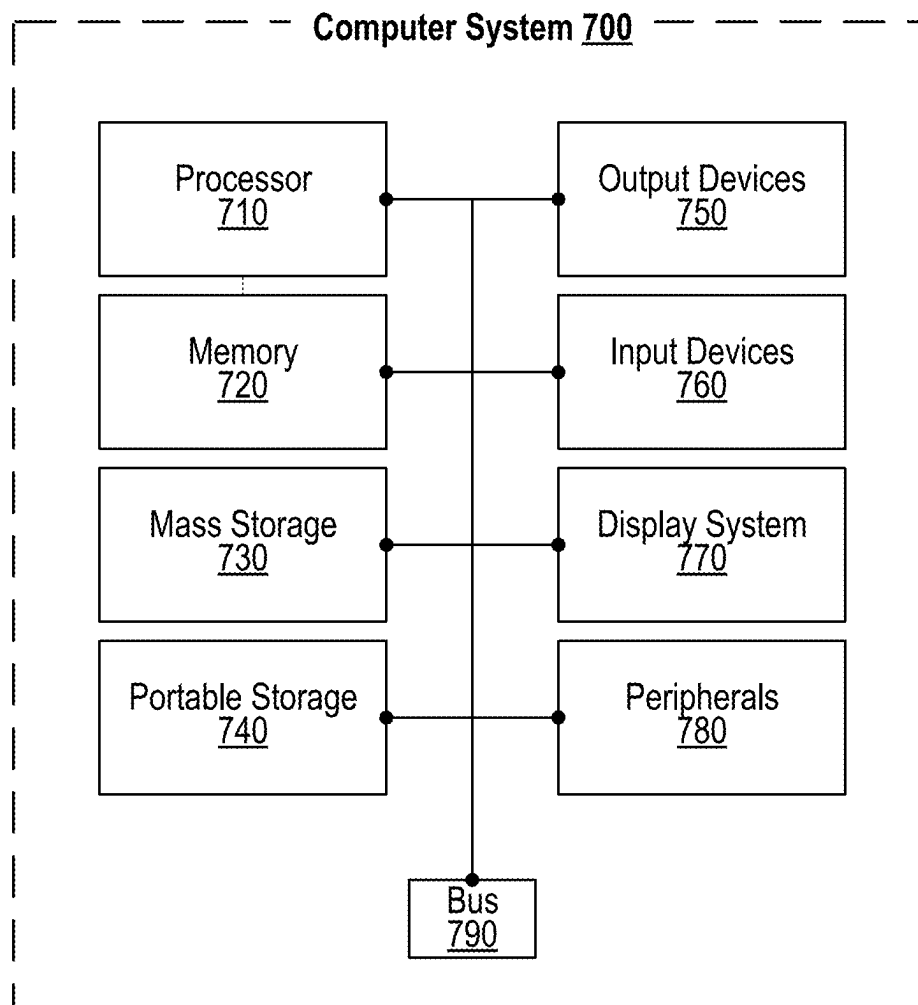


FIG. 7

1

AREA-BASED IDENTIFICATION OF DEVICES FOR PAIRING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/723,100, filed on Apr. 18, 2022, entitled, "MULTI-DEVICE AUTHORIZATION", which is a continuation of U.S. application Ser. No. 16/533,095, filed on Aug. 6, 2019, entitled, "MERCHANT POINT OF SALE COLLABORATING WITH PAYMENT READER TERMINAL VIA SERVER APPLICATION PROGRAMMING INTERFACE", now U.S. Pat. No. 11,328,277, and which are hereby expressly incorporated herein by reference in their entireties and for all purposes.

BACKGROUND

Payment cards, such as credit cards and debit cards, are often used by customers during transactions with merchants. Merchants can read payment information from payment cards using payment card reader terminals, for example via magnetic stripe readers or Europay/Mastercard/Visa (EMV) chip readers in the payment card reader terminals. Payment card reader terminals read payment card information from the payment card and send that payment card information to a server associated with a financial entity, such as a bank or credit card institution, in order to process the transaction by transferring funds from a customer account to a merchant account.

Merchant point of sale (POS) devices are systems that are used by merchants to enter items or services requested by a customer, retrieve prices for each item or service, calculate a total, and in some cases prepare a receipt or invoice to be printed and given to the customer before or after payment processing. Merchant POS systems can be customized for different industries. For example, some merchant POS systems are connected to bar code scanners that can scan bar codes attached to various items sold at the stores, so that items selected for purchase by the customer can quickly be identified to the merchant POS device.

A merchant POS and a payment terminal may sometimes be co-located in a single device. In some cases, however, a merchant may need to use a particular merchant POS device that is specially designed for the merchant's needs. A merchant may also wish to offload receipt of customer personal data to a separate payment terminal device to reduce the risk of loss of that customer personal data in a data breach. The benefits of using separate a merchant POS device and a separate payment card reader terminal thus include granting the merchant freedom to customize their merchant POS device without compromising payment security and functionality.

There is a need for a system that enables intuitive, efficient, and secure pairing and collaboration between a merchant POS device and a payment card reader terminal.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a system architecture including a merchant point of sale (POS) device, a payment reader terminal, a checkout service server, and a financial entity server.

FIG. 2 is a swim lane diagram illustrating a merchant point of sale (POS) device, a payment reader terminal, and a checkout service cooperating to process a transaction between a customer and a merchant.

2

FIG. 3 is a swim lane diagram illustrating authorized pairing of a merchant point of sale (POS) device and a payment reader terminal via a checkout service.

FIG. 4 is a swim lane diagram illustrating interactive customized messaging at the payment reader terminal.

FIG. 5 is an architecture diagram of a payment reader terminal device.

FIG. 6 illustrates a system architecture including a web-based point of sale (POS) device and mobile device that collaborate to perform a cardless transaction.

FIG. 7 is a block diagram of an exemplary computing device that may be used to implement some aspects of the technology.

DETAILED DESCRIPTION

A merchant uses a merchant point of sale (POS) device to identify items or services for purchase by a customer, and costs for each. One or more payment amounts are sent from the merchant POS device to a checkout server using an application programming interface (API), which the checkout server sends on to an authorized payment terminal device that then reads payment information from a payment object and sends the payment information back to the checkout server. The checkout server processes the transaction and confirms transaction completion to the merchant POS device and payment terminal device.

FIG. 1 illustrates a system architecture including a merchant point of sale (POS) device, a payment reader terminal, a checkout service server, and a financial entity server.

Payment instruments, also referred to as payment objects, transaction instruments, or transaction objects, may include payment cards or transaction cards such as credit cards, debit cards, gift cards, or transit cards. Payment instruments may also include payment devices or transaction devices, such as cellular phones, wearable devices, smartphones, tablet devices, laptops, media players, portable gaming consoles, and other computing devices 700 as discussed with respect to FIG. 7. Payment instruments may store payment instrument information (which may be referred to as payment object information, payment card information, payment device information, payment information, transaction object information, transaction card information, transaction device information, or transaction information), for example encoded along a magnetic stripe on the payment instrument, stored on an integrated circuit (IC) chip such as a Europay/Mastercard/Visa (EMV) chip, or stored in a non-transitory computer-readable storage medium electrically coupled to one or more active and/or passive near field communication (NFC) transceivers of the payment instrument.

The payment reader terminal device 125 of FIG. 1 includes hardware for reading payment instrument information from a one or more types of payment instrument. In particular, the payment reader terminal device 125 of FIG. 1 is illustrated with a slot on the right-hand side with a magnetic stripe reader. A payment card can be swiped through the slot, allowing the magnetic stripe reader of the payment reader terminal device 125 to read the payment instrument information from a magnetic stripe of the payment card. The payment reader terminal device 125 of FIG. 1 is also illustrated with a slot at the anterior end with an integrated circuit (IC) chip reader, such as a Europay/Mastercard/Visa (EMV) chip reader. One end of a payment card can be inserted into the slot, allowing the IC chip reader to read the payment instrument information from an IC chip of the payment card, which may be an EMV chip of the payment card. The payment reader terminal device 125 of

3

FIG. 1 also includes one or more active and/or passive near field communication (NFC) transceivers. A NFC-capable payment instrument can be placed adjacent to the one or more NFC transceivers of the payment reader terminal device 125, allowing the one or more active and/or passive NFC transceivers of the payment card terminal to read the payment instrument information from one or more active and/or passive NFC transceivers of the payment instrument. NFC-capable payment cards may include passive NFC transceivers, but may in some cases instead or additionally include active NFC transceivers. NFC-capable payment devices may include active NFC transceivers, but may in some cases instead or additionally include passive NFC transceivers.

The payment reader terminal device 125 of FIG. 1 also includes a display screen, which may include any type of display system 770 illustrated in and discussed with respect to FIG. 7. The payment reader terminal device 125 may render various messages or interfaces that for display on the display screen, such as messages with payment instructions instructing customers or merchants to swipe a payment card through a magnetic stripe reader slot of the payment reader terminal device 125, insert a payment card into an IC reader slot of the payment reader terminal device 125, or tap a NFC-capable payment instrument adjacent to one or more NFC transceivers of the payment reader terminal device 125. The payment reader terminal device 125 may in some cases render a keypad or keyboard for display on the display screen, so that the customer may enter a personal identification number (PIN), a password, a phone number, a member number, an email address, a signature, or some other customer identifier code via a touch-sensitive interface of the display screen. As discussed with respect to FIG. 4, the display screen may also receive additional messages from the checkout service 110, which may for example include advertisements or offers for additional items or services to purchase from the merchant, or coupons or discounts that may be applied, and which may be rendered by the payment reader terminal device 125 and displayed by its display screen until the payment reader terminal device 125 receives an order for a transaction to process, which may simply include a payment amount to charge the customer as discussed with respect to FIG. 2.

A merchant point of sale (POS) device 120 is also illustrated in FIG. 1. The merchant POS device 120 is a system that is used by a merchant to identify one or more purchases from the merchant by a customer (i.e., one or more items or services requested for purchase from the merchant by a customer), retrieve prices for each identified item or service, calculate a total, and in some cases prepare a receipt or invoice to be printed and given to the customer before or after payment processing. In some cases, the items or services may be identified by selecting an icon or text box corresponding to the item or service via an input device of the merchant point of sale (POS) device 120, such as a touch-sensitive surface of a touchscreen, a keypad, a keyboard, a mouse or trackpad controlling a mouse cursor, any other input device. In some cases, the items or services may be identified using a bar code scanner in communication with the merchant POS device 120, the merchant using the bar code scanner to scan bar codes affixed to the items, packaging associated with the items, tickets or vouchers associated with the services, or some combination thereof. In some cases, an amount or quantity of one or more of the items may be determined based on weight as determined by a scale in communication with the merchant POS device 120, the merchant using the scale to weigh the items (e.g.,

4

fruits, vegetables, grains, meats, or other consumables). In some cases, the merchant POS device 120 may handle appointments or reservations for services such as restaurant meals or doctor's appointments. The hardware and/or software of merchant POS devices 120 and used with merchant POS devices 120 may be customized differently for different industries. The merchant POS device 120 of FIG. 1 is illustrated as a touchscreen device showing an item or service identified "AAA" costing \$20.00 as a first purchase and an item or service identified "BBB" costing \$12.99 as a second purchase, the first purchase and second purchase summed up by the merchant POS devices 120 to a total payment amount of \$32.99.

The merchant POS device 120 and the payment reader terminal device 125 of FIG. 1 are both located at a merchant location 105, but are not directly connected to one another through any wired means nor through any wireless means, though in some cases they may be connected to the same Wi-Fi network, other WLAN network, or cellular phone network. Instead, the merchant POS device 120 and payment reader terminal device 125 are both connected through network connections to one or more checkout service servers 130 of the checkout service 110. In particular, the network connection(s) between the merchant POS device 120 and the one or more checkout service servers 130 are illustrated via arrow lines 140 and 145 in FIG. 1, while the network connection(s) between the payment reader terminal device 125 and the one or more checkout service servers 130 are illustrated via arrow lines 150 and 155 in FIG. 1. The one or more checkout service servers 130 may connect via network connection(s) to one or more financial entity servers 135 associated with one or more financial entities 115, which may be banks, credit card institutions, or other financial institutions, the network connection(s) illustrated via arrow lines 160 and 165 in FIG. 1. In some cases, the merchant POS device 120 and payment reader terminal device 125 may be connected in a wired or wireless manner, but not for the transfer of payment data or data related to the transaction (e.g., instead to share a power supply or to transfer other non-sensitive information).

The checkout service 110 may include an application programming interface (API) 170 through which the merchant POS device 120 and/or the payment reader terminal device 125 may communicate with the one or more checkout service servers 130. That is, the one or more checkout service servers 130 store include instructions that, when executed by one or more processors of the one or more checkout service servers 130, allow or cause the processors to interpret information sent to the one or more checkout service servers 130 from the merchant POS device 120 and/or the payment reader terminal device 125 when the information is generated by calling or invoking one or more functions of the API that in some cases format the information according to a format dictated by the API. Execution of the instructions by the one or more processors of the one or more checkout service servers 130 may further allow or cause the processors to format information according to the format dictated by the API before sending the information from the one or more checkout service servers 130 to the merchant POS device 120 and/or the payment reader terminal device 125. The information (in either direction) may include, for example, GET or HEAD actions requesting data, PUT or POST or PATCH actions creating and/or updating and/or overwriting and/or modifying and/or sending data, DELETE actions deleting data, CONNECT actions establishing connections between two or more devices, OPTIONS actions requesting data about available commu-

5

nication options, TRACE actions requesting a response, other HTTP methods, or some combination thereof. In some cases, the information (in either direction) may include, for example, payment requests, requests for further information, database queries, database query results, instructions or commands, payment instrument information, a payment amount update or modification, a transaction update or modification, a transaction cancellation, a merchant POS device identifier, a payment reader terminal device identifier, another device identifier, other messages or types of information, or some combination thereof. In some cases, the API may be a Representation State Transfer (RESTful) API. In some cases, the format dictated by the API may be that of an HTTP request or a HTTP response.

One example of the communications **140**, **145**, **150**, **155**, **160**, and **165** is given below. In this example, the arrow line **140** represents the merchant POS device **120** sending a payment amount and a merchant POS device identifier (ID) to the checkout service **110** (i.e., to the one or more checkout service servers **130**), optionally by calling the API **170** from the merchant POS device **120**. The arrow line **150** represents the payment reader terminal device **125** sending a payment terminal device identifier (ID) to the checkout service **110**, optionally by calling the API **170** from the payment reader terminal device **125**. The arrow line **150** may also represent the payment reader terminal device **125** querying or polling the checkout service **110** as to the existence of any transactions or orders pending processing. The arrow line **155** may represent the checkout service **110** sending the payment amount to the payment reader terminal device **125**, optionally along with a request to the payment reader terminal device **125** to initiate reading of payment instrument information, optionally calling the API **170** from the checkout service **110**. The arrow line **150** may then represent the payment reader terminal device **125** sending the payment instrument information (optionally encrypted at the payment reader terminal device **125**) to the checkout service **110** after the payment reader terminal device **125** reads the payment instrument, optionally by calling the API **170** from the payment reader terminal device **125**.

The arrow line **160** represents the checkout service **110** sending at least the payment instrument information and the payment amount (and optionally information identifying the customer, an account associated with the customer, the merchant, an account associated with the merchant, or a unique identifier associated with any of these) to the one or more financial entity servers **135** associated with the one or more financial entities **115**. Based on the information that the financial entity servers **135** receive from arrow **160**, the one or more financial entity servers **135** associated with the financial entities **115** identify an account associated with the customer, such as the customer's personal bank account, an account of an organization that the customer belongs to, or a credit account associated with the customer's credit card or a credit card of an organization that the customer belongs to. The one or more financial entity servers **135** associated with the financial entities **115** identify an account associated with the merchant, and process a transfer of funds in an amount equivalent to at least the payment amount (optionally also including one or more added tips, taxes, and/or fees) from the identified account associated with the customer to the identified account associated with the merchant. The arrow line **165** represents the one or more financial entity servers **135** associated with the financial entities **115** confirming to the checkout service **110** whether the transfer was success-

6

ful, meaning the transaction was successfully processed, or unsuccessful, meaning the transaction processing was unsuccessful.

The arrow line **145** represents the checkout service **110** confirming whether the transaction was successfully processed to the merchant POS device **120**, optionally via the API **170**. The arrow line **155** represents the checkout service **110** confirming whether the transaction was successfully processed to the payment reader terminal device **125**, optionally via the API **170**. Upon receiving the confirmations, the merchant POS device **120** and/or payment reader terminal device **125** may return to "default" or "end of transaction" or "next transaction" screens or interfaces in which they are ready for a next transaction. In some cases, the arrow line **145** or **155** may also include receipt data generated at the checkout service **110** and used by the merchant POS device **120** and/or the payment reader terminal device **125** to print a paper receipt using a printer of the merchant POS device **120** and/or the payment reader terminal device **125**. The receipt may alternately or additionally be sent by the checkout service **110** (and/or by the merchant POS device **120** and/or by the payment reader terminal device **125**) to a phone number (via text message), email address, or messaging platform username associated with the customer.

In some cases, the checkout service **110** and/or payment reader terminal device **125** may perform a cardless transaction. That is, the merchant POS device **120** may receive a customer identifier from the customer, for example as entered by the merchant or the customer via a user interface of the merchant POS device **120**. The customer identifier (customer ID) may be, for example, the customer's name, phone number, email address, mailing address, bank account number, credit account number, driver's license number, government-issued identification number, username, personal identification number (PIN), a unique code, some other identifier establishing the customer's identity, or some combination thereof. The merchant POS device **120** may send the customer identifier, and optionally also a merchant identifier (merchant ID) similarly establishing the identity of the merchant, to the checkout service **110** along arrow line **140** along with the payment amount. The checkout service **110** may then query a database for the customer identifier to locate an account at the checkout service associated with the customer, query the database for the merchant identifier to locate an account at the checkout service associated with the merchant, and perform a transfer of at least the payment amount from the account associated with the customer to the account associated with the merchant. In some cases, the customer identifier and/or the merchant identifier are sent by the checkout service **110** to the payment reader terminal device **125** or to the one or more financial entity servers **135**, so that the database queries and transfer of at least the payment amount from the account associated with the customer to the account associated with the merchant can occur at the payment reader terminal device **125**, the one or more checkout service servers **130**, the one or more financial entity servers **135**, or some combination thereof. Confirmations upon completion of the transfer may then be sent to the payment reader terminal device **125** and/or to the merchant POS device **120** by the checkout service **110**.

A merchant POS and a payment terminal may sometimes be co-located in a single device. In some cases, however, a merchant may need to use a particular merchant POS device that is specially designed for the merchant's needs, for example to connect to a particular inventory management system, or to connect to specialized hardware such as a bar

code scanner or a scale to measure weight of items being purchased. A merchant may also wish to offload receipt of customer personal data to a separate payment terminal device to reduce the risk of loss of that customer personal data in a data breach. The benefits of using separate a merchant POS device and a separate payment card reader terminal thus include granting the merchant freedom to customize their merchant POS device without compromising payment security and functionality.

The merchant POS device **120**, the payment reader terminal device **125**, the one or more checkout service servers **130**, and the one or more financial entity servers **135** may each include one or more computing devices **700**, or may include at least a subset of the components of the computing device **700** that are illustrated in and/or discussed with respect to FIG. 7.

In some cases, the merchant POS device **120** may be identified herein as the merchant device or the POS device. In some cases, the payment reader terminal device may be referred to herein as the payment reader, the payment card reader, the payment instrument reader, the payment object reader, the payment reader terminal, the payment card reader terminal, the payment instrument reader terminal, the payment object reader terminal, the payment terminal, the payment terminal device, the terminal device, the payment object reader device, the payment reader device, the payment card reader device, the payment instrument reader device, the payment object reader device, any of the terms identified above with the word “payment” substituted with the word “transaction,” or some combination thereof. In some cases, the one or more checkout service servers **130** may be referred to collectively herein as the checkout service **110**. In some cases, the one or more financial entity service servers **130** may be referred to collectively herein as the one or more financial entities **115** or as the one or more payment processing services.

FIG. 2 is a swim lane diagram illustrating a merchant point of sale (POS) device, a payment reader terminal, and a checkout service cooperating to process a transaction between a customer and a merchant. In particular, the operations **200** of FIG. 2 are illustrated as being performed by the merchant POS device **120**, the payment reader terminal device **125**, the checkout service server(s) **130**, the financial entity server(s) **135**, or some combination thereof.

At step **205**, the merchant POS device **120** sends a request to process a payment amount for a transaction between a customer and a merchant to the checkout service, the payment amount corresponding to one or more purchases by the customer from the merchant, the request formatted for an API **170** of the checkout service **110** and optionally including a merchant POS device identifier (merchant POS device ID). At step **210**, the checkout service server(s) **130** receive the request to process the payment amount from the merchant POS device **120**, and optionally receive the merchant identifier. In some cases, the merchant POS device **125** may also send a customer identifier identifying the customer and/or a merchant identifier identifying the merchant to the checkout service **110**, which the checkout service **110** may receive and optionally use to identify the customer and/or the merchant, and/or send on to the one or more financial entity servers **135** (e.g., at step **245**) so that the financial entity servers **135** can use the customer identifier and/or merchant identifiers to identify the customer and/or the merchant.

At step **215**, the checkout service server(s) **130** identify a payment reader terminal device **125** that is authorized to cooperate with the merchant POS device **120**. In some cases,

the checkout service server(s) **130** performs this by querying a database using the merchant POS identifier sent by the merchant POS device **120** at step **205** and received at the checkout service **110** at step **210**, and retrieving a record from the database as an output of the query. The record from the database may identify the merchant POS identifier as well as one or more payment reader terminal device identifiers that identify one or more payment reader terminal devices that are authorized to cooperate with the merchant POS device **120** identified by the merchant POS identifier. The payment reader terminal device identifier that identifies the payment reader terminal device **125** discussed in FIG. 2 in particular is identified as authorized to cooperate with the merchant POS device **125** based on the payment reader terminal device identifier being one of the one or more payment reader terminal device identifiers in the record. The record may be generated during step **350** of a pairing process **300** discussed in FIG. 3.

At step **220**, the checkout service server(s) **130** transmit the payment amount from the checkout service to the payment reader terminal device **125**. At step **225**, the payment reader terminal device **125** receives the payment amount from the checkout service server(s) **130**.

At step **230**, the payment reader terminal device **125** reads payment instrument information from a payment instrument, for example from a magnetic stripe or an IC chip of a payment card, or from a NFC transceiver of a NFC-capable payment instrument. At step **235**, the payment reader terminal device **125** transmits the payment card information and optionally a payment reader terminal identifier to the checkout service **110** formatted for the API **170** of the checkout service **110**. At step **240**, the checkout service server(s) **130** receive the payment card information at the checkout service from the payment reader terminal device **125**, optionally along with the payment reader terminal identifier.

At step **245**, the checkout service server(s) **130** trigger processing of a transfer of at least the payment amount from an account of the customer to an account of the merchant by transmitting at least the payment card information and the payment amount from the checkout service to financial entity. In some cases, the checkout service server(s) **130** may add a tip, taxes, fees, and/or other additional amounts to the payment amount if they have not already been added. In other cases, the tip, taxes, fees, and/or other additional amount may have been added to the payment amount before step **245** at the merchant POS device **120** or at the payment reader terminal device **125**. At step **250**, the financial entity server(s) **135** receive the payment instrument information and the payment amount from the checkout service **110**. At step **252**, the financial entity server(s) **135** transfer at least the payment amount from an account of the customer to an account of the merchant **250**.

In some cases, the payment instrument information is encrypted using an encryption key by the payment reader terminal device **125** after it is read at step **230** but before it is transmitted at step **235**, and is decrypted by the financial entity server(s) **135** using a corresponding encryption key. The encryption key stored at the payment reader terminal device **125** and the corresponding encryption key stored at the financial entity server(s) **135** may be the same encryption key, for example if symmetric encryption is used. The encryption key stored at the payment reader terminal device **125** and the corresponding encryption key stored at the financial entity server(s) **135** may be different encryption keys, for example if asymmetric encryption is used. For example, if a public key infrastructure (PKI) is used, one of

the keys may be a private key while the other is a public key. The payment instrument information may also be hashed and/or tokenized at the payment reader terminal device 125 and/or at the financial entity server(s) 135. Encryption algorithms that may be used may include, for example, Rivest-Shamir-Adleman (RSA) encryption, Secure Hash Algorithm (SHA), Elliptic Curve Cryptography (ECC), Diffie-Hellman, or some combination thereof.

At step 255, the financial entity server(s) 135 send a confirmation that the transfer has been completed successfully, and therefore that the transaction has been processed successfully, to the checkout service 110. At step 260, the checkout service server(s) 130 receive this confirmation that the transfer has been completed successfully, and therefore that the transaction has been processed successfully, from the financial entity server(s) 135. At step 265, the checkout service server(s) 130 send a confirmation that the transfer has been completed successfully, and therefore that the transaction has been processed successfully, to the merchant POS device 120 and/or to the payment reader terminal device 125. At step 270, the merchant POS device 120 and/or to the payment reader terminal device 125 receive the confirmation that the transfer has been completed successfully, and therefore that the transaction has been processed successfully, from the checkout service server(s) 130. Upon receiving the confirmation, the merchant POS device 120 and/or to the payment reader terminal device 125 can reset to a default mode in which the merchant POS device 120 and/or the payment reader terminal device 125 can prepare for a next transaction.

In some cases, the checkout service server(s) 130 also generate a receipt at step 260, and the confirmation(s) sent at step 265 include receipt data to be used by the merchant POS device 120 and/or the payment reader terminal device 125 for printing, displaying, or sending the receipt. For example, the merchant POS device 120 and/or the payment reader terminal device 125 may print out the receipt using a printer of the merchant POS device 120 and/or of the payment reader terminal device 125. The merchant POS device 120 and/or the payment reader terminal device 125 may request input of a phone number or email address or messenger platform username of the customer (unless the phone number or email address or messenger platform username has already been received by the merchant POS device 120 and/or the payment reader terminal device 125) and may send the receipt to the customer over text message or email or messenger platform. The merchant POS device 120 and/or the payment reader terminal device 125 may send the phone number or email address or messenger platform username to the checkout service 110 (if it has not sent this information already, or if the checkout service 110 does not have this information already such as from a previous transaction) so that the checkout service 110 may send the receipt to the user over text message or email or messenger platform.

In some cases, the checkout service server(s) 130 may generate an "order" data structure based on the payment amount upon receipt of the request at step 210. The order data structure may include, for example, the payment amount, the merchant identifier, and the customer identifier. In some cases, the order may include a list of the one or more purchases requested by the customer—that is, the one or more items and/or services that the customer is requesting to purchase—and optionally prices for each of these. In some cases, the total payment amount may be calculated by the service server(s) 130 rather than by the merchant POS device 120. Step 220 may in some cases include transmitting

the payment amount, but it may instead include transmitting the order data structure, or at least a subset of the information stored in the order data structure. In some cases, thus, the checkout service server(s) 130 send information identifying the one or more purchases requested by the user, and/or the prices associated with each, to the payment reader terminal device 120 at step 220, and the total payment amount may be calculated by the payment reader terminal device 120 rather than by the checkout service server(s) 130 or by the merchant POS device 120. In some cases, an order may be updated or modified before the payment instrument is read, or even after the payment instrument is read but before the transaction is processed at the financial entity server(s) 135. In such cases, the merchant POS device 120 or payment reader terminal device 125 can send a request to the checkout services server(s) 130 to update the order data structure, and the checkout services server(s) 130 can notify the merchant POS device 120 and/or the payment reader terminal device 125 about the update or modification to the order if necessary, can update the order data structure, can update the total payment amount if applicable, and can perform step 220 again by sending the updated order or payment amount to the payment reader terminal device 125 for payment instrument reading and/or can perform step 245 to trigger processing of the transaction with the updated order and/or payment amount.

While the operations 200 of FIG. 2 do not illustrate entry of a PIN, a signature, selection of a tip amount, selection of a donation amount to a charity, or any such customer information entry action, it should be understood that any of these may occur at the merchant POS device 120 and/or at the payment reader terminal device 125, in some cases depending on the capabilities of each. If the payment reader terminal device 125 includes a display screen as illustrated in FIG. 1, as well as a touchscreen interface or a keypad, then these can be received at the payment reader terminal device 125, and the payment reader terminal device 125 can send a request to the checkout service 110 to update or modify the order or payment amount if appropriate, or to generate a new transaction if appropriate. If the payment reader terminal device 125 does not include a display screen or an appropriate touchscreen interface or keypad, or if the merchant chooses not to use these, then these can be received at the merchant POS device 120, and the merchant POS device 120 can send a request to the checkout service 110 to update or modify the order or payment amount if appropriate, or to generate a new transaction if appropriate. In the case of a donation to a charity, a new transaction may be requested, since the donation amount will be paid to the charity rather than to the merchant, with the new order data structure for this new transaction including information identifying the charity so that the financial entity server(s) 135 can be notified by the checkout service server(s) 130 to transfer the customer's funds to an account associated with the charity.

In some cases, the merchant POS device 120 may be a web-based POS system on a website rather than a device physically located at the merchant location 105. In this case, the merchant POS device 120 may be a web server that serves the website, a browser device that browses the website, or some combination thereof. The payment reader terminal device 125 may, in this situation, be a device that enabled a cardless transaction, such as the customer's own mobile device. Such a situation is illustrated in and discussed further with respect to FIG. 6.

In some cases, after the operations 200 of FIG. 2 have been performed one or more times for one or more separate

11

transaction involving one or more customers, the checkout service server(s) 130 can generate anonymized analytics based on data collected during each of the transactions, and send the anonymized analytics back to the merchant POS device 120 or to another computing device associated with the merchant. The data can be anonymized by stripping customer identifiers, such as names, phone numbers, member numbers, addresses, from the data, or by generating new customer identifiers (e.g., unique strings of alphanumeric characters) that cannot be traced to the original customer identifiers. The anonymized analytics may, for example, show a merchant peak purchasing times, show a merchant slow times in which the fewest purchases are made, or show a merchant how many of its customers are purchasing a particular item or service. In some cases, a wireless beacon or wireless network may be used to detect mobile devices of persons entering the merchant's storefront (or vicinity), in which case the anonymized analytics may show how many of the persons entering the merchant's storefront (or vicinity) become customers. In some cases, these analytics may not be anonymized, though anonymization can be valuable in that, if the merchant suffers a data breach, personal customer data, such as any of the customer identifiers discussed herein, are not disclosed.

FIG. 3 is a swim lane diagram illustrating authorized pairing of a merchant point of sale (POS) device and a payment reader terminal via a checkout service. In particular, the operations 300 of FIG. 3 are illustrated as being performed by the merchant POS device 120, the payment reader terminal device 125, the checkout service server(s) 130, the financial entity server(s) 135, or some combination thereof.

At step 305, the checkout service server(s) 130 optionally provide an API key of the API 170 of the checkout service 110 to developers of the merchant POS device 120 so that the API key can be used in the code run by the merchant POS device 120. At step 310, developers customize the software run by the merchant POS device 120 to call the API 170 of the checkout service, optionally using the API key sent at step 305.

At step 315, the merchant POS device 120 sends a merchant POS device identifier (ID) to the checkout service 110, optionally formatted for the API 170. At step 320, the checkout service server(s) 130 receive the merchant POS device identifier (ID) from the merchant POS device 120. In some cases, the merchant POS device identifier may be a serial number or model number of the merchant POS device 120, or another unique identifier of the merchant POS device 120.

At step 325, the payment reader terminal device 125 sends a payment reader terminal device identifier (ID) to the checkout service 110, optionally formatted for the API 170. At step 330, the checkout service server(s) 130 receive the merchant POS device identifier (ID) from the merchant POS device 120. In some cases, the payment reader terminal device identifier may be a serial number or model number of the payment reader terminal device 125, or another unique identifier of the payment reader terminal device 125. In some cases, step 325 is performed as a result of the payment reader terminal device 125 receiving an input via an interface (e.g., the touchscreen) of the payment reader terminal device 125, the input optionally being from the merchant, the input optionally agreeing to pair the payment reader terminal device 125 with the merchant POS device 120. In some cases, the input from the merchant is a login from the merchant or an entry of a merchant identifier of the merchant into the payment reader terminal device 125, in which case

12

payment reader terminal device 125 may also send the merchant identifier or merchant login to the checkout service 110 along with the terminal device identifier at step 325 to simultaneously act as the notification of step 340. Steps 325 and 330 may occur before, after, or during any of steps 305, 310, 315, and/or 320. In some cases, steps 325 and 330 may not occur at all, and the payment reader terminal device identifier may be sent to the checkout service 110 for the first time at steps 340 and 345.

At step 335, the payment reader terminal device identifier is optionally input into the merchant POS device 120. In some cases, the payment reader terminal device identifier may be punched in via a keypad, keyboard, touchscreen, or other input interface. In other cases, the payment reader terminal device identifier may be scanned in to the merchant POS device 120 via scanning of a bar code or quick response (QR) code that is on the housing of the payment reader terminal device 125 or that is displayed on a display screen of the payment reader terminal device 125. At step 340, the merchant POS device 120 optionally notifies the checkout service 110 that the payment reader terminal device 125 was identified to the merchant POS device 125, optionally via the API 170. At step 345, the checkout service server(s) 130 receive the notification, optionally from the merchant POS device 120. The notification of steps 340 and 345 may optionally include the payment reader terminal device identifier as input or otherwise received by the merchant POS device 120. In some cases, a different device other than the merchant POS device 120 may perform steps 335 and 340, such as a personal user device (e.g., a smartphone) associated with the merchant, in which case it is a notification that the payment reader terminal device 125 has been identified to the merchant generally more so than to the merchant POS device 120 specifically. Use of the different device may increase flexibility by reducing how much the merchant POS device 125 must be customized at step 310, and may make pairing easier by using a device that the merchant is already familiar with.

In some cases, receipt at the checkout service 110 of the notification indicating that the terminal device has been identified to the merchant at step 345 may include receipt of a serial number (or other payment reader terminal device identifier) associated with the payment reader terminal device 125 from a device other than the payment reader terminal device 125, such as the merchant POS device 120 or mobile device of the merchant. In some cases, receipt at the checkout service 110 of the notification indicating that the terminal device has been identified to the merchant at step 345 may include receipt of a code generated by and output by the payment reader terminal device 125 (e.g., the code output via a display and/or speaker of the payment reader terminal device 125) from a device other than the payment reader terminal device 125, such as the merchant POS device 120 or mobile device of the merchant. In some cases, receipt at the checkout service 110 of the notification indicating that the terminal device has been identified to the merchant at step 345 may include receipt of a code output by a mobile device in response to the mobile device interacting with the payment reader terminal device 125 (e.g., in response to the mobile device scanning a QR code displayed by the payment reader terminal device 125) from a device other than the payment reader terminal device 125, such as the merchant POS device 120 or the mobile device. In some cases, receipt at the checkout service 110 of the notification indicating that the terminal device has been identified to the merchant at step 345 may include receipt of an indication that the payment reader terminal device 125 (or an identifier

13

of the payment reader terminal device 125) has been selected from a list of devices (or identifiers thereof) at the merchant device. In some cases, receipt at the checkout service 110 of the notification indicating that the terminal device has been identified to the merchant at step 345 may include receipt of a notification from the merchant POS device 120 indicating that the merchant POS device 120 has received a payment reader terminal device identifier (e.g., serial number, code generated by the payment reader terminal device) associated with the payment reader terminal device 125. In some cases, receipt at the checkout service 110 of the notification indicating that the terminal device has been identified to the merchant at step 345 may include receipt of a notification from the payment reader terminal device 125 including a merchant identifier in response to receiving a merchant input at the payment reader terminal device 125 with the merchant identifier.

At step 350, in response to receipt of the notification from the merchant POS device 120 or from the different device at step 345, the checkout service server(s) 130 generates a pairing record in a database that is stored at or otherwise accessible to the checkout service server(s) 130, the pairing record identifying at least the merchant POS device identifier received in step 320 and the payment reader terminal device identifier received in step 330 (and/or in step 345). The pairing record indicates that the payment reader terminal device 125 corresponding to the payment reader terminal device identifier is authorized to collaborate with the merchant POS device 120 corresponding to the merchant POS device identifier in that the payment reader terminal device 125 is authorized to read payment instrument information for use in processing one or more transactions initiated by the merchant POS device 120. The pairing record of step 350 may be the same record discussed with respect to step 215 of the operations 200 of FIG. 2.

In some cases, more than one payment reader terminal device 125 may be paired with a particular merchant POS device 120. In this case, the merchant POS device 120 may present the merchant with a list of authorized payment reader terminal devices 125 that the merchant can use with the particular merchant POS device 120 he or she is using, from which the merchant can select one via a user interface of the merchant POS device 120. This may be useful in situations where a merchant wishes to move about different locations within their a storefront with only the merchant POS device 120, which allows the merchant to then use the nearest payment reader terminal device 125. To facilitate this use case, a map may be used instead of a list, the map optionally showing both the locations of each of the paired/authorized payment reader terminal devices 125 as well as the location of the merchant POS device 120 (and therefore of the merchant). The locations of each of these devices may be identified based on positioning receivers of the payment reader terminal devices 125 and/or of the merchant POS device 120, such as GNSS receivers, WLAN receivers, cellular network receivers, Bluetooth receivers, and the like. Bluetooth beacons may be distributed throughout the merchant location 105 (i.e., the storefront) to enhance indoor positioning, since GNSS-based typically works more accurately outdoors than indoors.

FIG. 4 is a swim lane diagram illustrating interactive customized messaging at the payment reader terminal. In particular, the operations 400 of FIG. 4 are illustrated as being performed by the merchant POS device 120, the payment reader terminal device 125, the checkout service server(s) 130, the financial entity server(s) 135, or some combination thereof.

14

At step 405, the checkout service server(s) 130 receive a message. At step 410, the checkout service server(s) 130 send the message the payment reader terminal device 125. At step 415, the payment reader terminal device 125 receives the message from the checkout service server(s) 130. In some cases, the checkout service server(s) 130 send instructions along with the message to the payment reader terminal device 125, the instructions identifying that the payment reader terminal device 125 should display the message until it is time for the payment reader terminal device 125 to read payment instrument information and/or otherwise fulfill its role in payment processing for the transaction initiated by the merchant POS device 120. The message may include text, one or more images, one or more videos, audio, or some combination thereof. The message may include a logo of the merchant, for example. The message may include offers, promotions, coupons, or discounts that may be applied to items or services already being purchased by the customer. The message may include one or more advertisements, offers, promotions, coupons, or discounts for additional items or services to purchase from the merchant, for example for items or services that the customer has purchased in the past, for items or services that are complementary or related to items or services that the customer is currently purchasing or has purchased in the past, or for items or services that are often purchased by other customers along with items or services that the customer is currently purchasing or has purchased in the past. Complementary or related to items or services may include items or services from the same brand, same manufacturer, same distributor, same category (e.g., office supplies, clothing, consumables), that are useful alongside one another (e.g., pen and paper and pencil, printer and paper and ink cartridge, meal and side dish and beverage, sun hat and sunglasses and sunscreen and swimwear and beach towel, warm hat and coat and gloves).

At step 420, a processor of the payment reader terminal device 125 renders the message for display, then displays the message via a display screen of the payment reader terminal device 125, at least until receipt of the payment amount by the payment reader terminal device 125 at step 220. At step 425, the payment reader terminal device 125 optionally receives an input from the customer via a user interface of the payment reader terminal device 125 (e.g., a touch-sensitive interface of a touch screen, a button) while the message is displayed via the display screen of the payment reader terminal device 125. At step 430, in response to receipt of the input at step 425, the payment reader terminal device 125 sends a request to the checkout service 110 for a modification to the transaction. The modification may depend on the content of the message. For example, if the message advertises an additional item or service, the modification may include adding the additional item or service for purchase as part of the transaction. If the message includes an offer, promotion, discount, or coupon applicable for a particular item or service, the modification may include reducing the price associated with the particular item or service, and as a result reducing the payment amount.

At step 435, the checkout service server(s) 130 receives the request for the modification to the transaction. At step 440, the checkout service server(s) 130 notifies the merchant POS device 120 regarding the request for the modification to the transaction, for example by sending the request for the modification to the transaction on to the merchant POS device 120, or by otherwise identifying the requested modification to the transaction to the merchant POS device 120. At step 445, the merchant POS device 120 receives the

15

notification sent at step 440. At step 450, the merchant POS device 120 modifies the transaction as requested, for example by adding the additional item or service, or by reducing the price of an item or service, and modifies the payment amount as needed if applicable.

Step 455 is identical to step 205 from the operations 200 of FIG. 2, though the request of step 455 has the updated payment amount associated with the transaction having the modification applied in step 450. Step 460 is identical to step 210 from the operations 200 of FIG. 2, though again with the new modified payment amount. Step 465 is identical to step 215 from the operations 200 of FIG. 2, though again with the new modified payment amount. Step 465 is identical to step 220 from the operations 200 of FIG. 2, though again with the new modified payment amount. Step 470 is identical to step 225 from the operations 200 of FIG. 2, though again with the new modified payment amount.

At step 475, in response to receipt of the payment amount at step 470, the payment reader terminal device 125 displays payment instructions via the display screen of the payment reader terminal device 125 until the payment instrument is received (e.g., into a slot of the payment reader terminal device 125) and/or read by one of the one or more payment instrument readers of the payment reader terminal device 125.

FIG. 5 is an architecture diagram of a payment reader terminal device.

The architecture diagram of the payment reader terminal device 125 of FIG. 5 includes a main processor 505 in a housing 590, and optionally one or more secure processor(s) 510 within a secure internal enclosure 595 within the housing 590. The main processor 505 may execute instructions to perform a variety of tasks, which may include communicating with the one or more secure processor(s) 510 as well as communicating with and/or controlling the various components coupled directly or indirectly to the main processor 505.

The main processor 505 is coupled to a main battery 540 and optionally a backup battery 545 in FIG. 5. In some cases, the main battery 540 may be rechargeable, while the backup battery 545 may be non-rechargeable. Ultimately, however, either or both of the batteries may be rechargeable or non-rechargeable. The main processor 505, main battery 540, and/or backup battery 545 may also be connected to a power connector (not pictured) that can be connected to an external battery (not pictured), an external generator (not pictured), or a power outlet (not pictured). The power connector can be used to recharge the main battery 540 and/or backup battery 545. The power connector can also be used to directly power the processor 505 and all other components of the payment reader terminal device 125, even if both the main battery 540 and backup battery 545 are missing.

The main processor 505 is coupled to a printer 570 via a printer controller 560. The printer 570 may be used to print receipts, coupons, barcodes, quick-response ("QR") codes, or some combination thereof. The printer 570 may be a thermal printer, a direct thermal printer, a thermal transfer thermal printer, a solid-ink "phaser" printer, an inkjet printer, a thermal inkjet ("bubblejet") printer, a continuous inkjet printer, a piezoelectric inkjet printer, a dye-sublimation printer, a laser printer, an LED printer, or some combination thereof. The printer controller 560 may include a memory and/or a processor, microcontroller or ASIC (application specific integrated controller) with which it may adjust

16

various printer setting values assigned to various printer settings to control various aspects of the how the printer 570 prints.

In some cases, the main processor 505 is coupled to a temperature sensor 575 in FIG. 5 used to ensure that the payment reader terminal device 125 stays within a temperature range in which it can operate safely and securely. The temperature sensor may optionally include one or more thermistors, and may optionally be connected to the main battery 540 and/or to the printer 570. The temperature sensor may include one or more of a Negative Temperature Coefficient (NTC) thermistor, a Resistance Temperature Detector (RTD), a resistance thermometer, a thermocouple, an infrared sensor, a bimetallic device, a thermometer, a change-of-state sensor, a semiconductor-based temperature sensor, a silicon diode or other diode with temperature-sensitive voltage characteristics, or some combination thereof.

The main processor 505 may be connected to one or more antennas, including an 802.11 Wi-Fi antenna 555, a cellular phone network antenna 550, a Bluetooth® antenna, a Bluetooth® Low Energy (BLE) antenna, any other antennae or communication means discussed with respect to the output devices 750 or input devices 760 of FIG. 7, or some combination thereof. The main processor 505 may alternately be connected to other means to networking, such as a wired Ethernet port.

The main processor 505 may be any type of processor 710 identified with respect to FIG. 7, and may be in some cases be part of, or include, an application-specific integrated circuit (ASIC). While the main processor 505 and secure processor(s) 510 are illustrated as separate components in FIG. 5, the main processor 505 and secure processor(s) 510 may in some cases be combined into a single component. It should be understood that any actions described herein as being performed by the secure processor(s) 510 can alternately or additionally be performed by the main processor 505, and that any actions described herein as being performed by the main processor 505 can alternately or additionally be performed by the secure processor(s) 510.

The secure internal enclosure 595 can include tamper protection circuitry 530, such as tamper traces or grids running along one or more surfaces of the secure internal enclosure, allowing the secure processor(s) 510 to detect attempts to tamper with the secure internal enclosure 595 or any components within. For example, tamper protection circuitry 530 can detect attempts to open the secure internal enclosure 595, to drill into the secure internal enclosure 595, to modify the secure internal enclosure 595, or to flood the secure internal enclosure 595 with a conductive fluid, such as a conductive ink. The tamper protection circuitry 530 can do this by measuring voltages at one or more points along a circuit connected to the tamper traces or grids running along the surface(s) of the secure internal enclosure 595, allowing the tamper protection circuitry 530 to detect short circuits, breaks in the circuit, or other changes in voltage outside of ordinary threshold levels, which would typically indicate a tampering attempt. Detection of a tamper attempt could be used by the secure processor(s) 510 or main processor 505 to disable at least a subset of the functions or components of the payment reader terminal device 125. In some cases, one or more conductive tamper traces run along an interior surface of the secure internal enclosure 595, and the tamper protection circuitry 530 is configured to detect a change in voltage along the one or more conductive tamper traces, for example caused by a break in a conductive tamper trace caused by a tamper attempt, or caused by a first conductive tamper trace that is normally (i.e., when not tampered with)

17

at a first voltage being connected during a tamper attempt to a second conductive tamper trace that is normally (i.e., when not tampered with) at a second voltage.

The secure internal enclosure **595** and its included tamper protection circuitry **530** can be used to protect any circuitry that reads, stores, or otherwise conveys sensitive information, such as credit or debit card numbers, near field communication (NFC) signal data from a NFC object conveying transaction information, bank account information, other payment instrument information, bank account information, credit account information, customer identifiers, merchant identifiers, payment reader terminal device identifiers, other transaction information, customer biometric information, customer contact information, customer identity information, customer photographs or facial scans, Personal Identification Number (PIN) numbers or codes, handwritten signature scans, handwritten signatures entered via a touch-sensitive surface of the payment reader terminal device **125**, digital certificate signatures, symmetric encryption keys, asymmetric encryption keys, other types of encryption keys, and so forth.

Because information entered via a touch-sensitive screen surface interface **525** of a touchscreen **210** may include sensitive information, such as PIN codes or handwritten signatures entered via touch-sensitive surface, circuitry reading inputs of the touch-sensitive surface may in some cases be housed in the secure internal enclosure **595**, while the circuitry of the display screen **520** of the touchscreen **210** might not be. Likewise, the payment reader terminal device **125** may include other input devices **760** as discussed with respect to FIG. 7, such as physical keypads, whose circuitry might also be located in the secure internal enclosure **595** for the same reasons. The touch-sensitive screen surface interface **525** may use any type of display discussed in reference to the display system **770** of FIG. 7, and may use capacitive touch sensing, resistive touch sensing, inductive touch sensing, or some combination thereof. For example, the touch-sensitive screen surface interface **525** may utilize a Indium Tin Oxide (ITO) touch-sensitive layer or a printed ink touch-sensitive layer.

The secure processor(s) **510** are also connected to one or more payment instrument reader(s) **515**. The payment instrument reader(s) **515** may include a magnetic stripe reader **235**, an integrated circuit (IC) chip reader **225**, a near field communication (NFC) reader, or some combination thereof. The magnetic stripe reader **235** reads payment instrument data from a magnetic stripe of a payment instrument such as a payment card. The IC chip reader **225** reads payment instrument data from an IC chip of a payment instrument such as a payment card. The IC chip may be a chip following Europay/Mastercard/Visa ("EMV") standards, also known as an EMV chip. The NFC reader reads payment instrument data from wireless NFC signals received from a payment instrument such as a payment card or payment device. The payment instrument may be a payment card, such as a credit card or a debit card. The payment instrument may be any type of NFC-capable device, such as an active NFC tag, a passive NFC tag, or a computer system **500** as described with respect to FIG. 5. In some cases, multiple payment instrument readers **515** may share components—for example, the IC chip reader **225** and the NFC reader may share a contactless antenna. Once the payment instrument reader(s) **515** retrieve the payment instrument data from the payment instrument, the payment instrument reader(s) **515** send the payment instrument data to the secure processor(s) **510** and/or main processor **505**. The payment instrument reader(s) **515** or secure processor(s)

18

510 sometimes modify or encrypt the payment instrument data using an encryption key stored in a memory (not pictured) associated with the payment instrument reader(s) **515** and/or secure processor(s) **510** before sending it to the main processor **505**.

The main processor **505** may also be connected to a memory **535** that stores various instructions to be executed by the main processor, such as instructions corresponding to a financial transaction software application allowing the main processor **505** to receive transaction data read by the payment instrument reader(s) **515**, optionally via the secure processor(s) **510**, and to transmit that data to one or more financial transaction processing server(s), such as credit or debit card processing server(s) and/or bank server(s), thereby conducting a financial transaction between a merchant and a buyer whose transaction information was read via the payment instrument reader(s) **515**. The memory **535** may also store settings associated with the printer controller **560**, the display screen **520**, the touch-sensitive screen surface interface **525**, the Wi-Fi antenna **555**, the cellular antenna **550**, the NFC transceiver, and various other components of the payment reader terminal device **125**.

The main processor **505** may also be connected to at least one external connector **580**, which may be an externally-facing female port or male plug that may allow the POS device **510** to connect to other devices or power sources. In some cases, the external connector **580** may allow the payment reader terminal device **125** to connect to any other computing device **700**, or to any device including at least a subset of the components identified in FIG. 8 or discussed in its description. The payment reader terminal device **125** may be connected to one or more peripheral devices or hubs via the at least one external connector **580**. The payment reader terminal device **125** may in some cases act as a peripheral device to a host device connected via the at least one external connector **580**. In some cases, the external connector **580** may function as the power connector discussed above with respect to the batteries **540/145** and/or external power sources. In some cases, the external connector **580** may be a universal serial bus (USB) port/plug using any USB standard, including USB Type-A, USB Type-B, USB Type-C, USB 5.x, USB 2.x, USB 3.x, USB 4.x, Thunderbolt, USB mini, USB micro, USB On-The-Go (OTG), any other USB standard, or some combination thereof. In other cases, the external connector **580** may be an Apple® Lightning® port/plug, an Ethernet port/plug, a fiber optic port/plug, a proprietary wired port/plug. In other cases, the external connector **580** may be a wireless transceiver, such as one supporting a BLUETOOTH® wireless signal transfer, a BLUETOOTH® low energy (BLE) wireless signal transfer, a radio-frequency identification (RFID) wireless signal transfer, near-field communications (NFC) wireless signal transfer, 802.11 Wi-Fi wireless signal transfer, cellular data network wireless signal transfer, a radio wave signal transfer, a microwave signal transfer, an infrared signal transfer, a visible light signal transfer, an ultraviolet signal transfer, a wireless signal transfer along the electromagnetic spectrum, or some combination thereof.

The payment reader terminal device **125** may also include various additional or alternative elements connected to the main processor **505** and/or the secure processor(s) **510**, each of which may optionally be located in the secure internal enclosure **595**. Such additional or alternative elements may include microphones. The additional or alternative elements may include cameras, such as visible light cameras, infrared cameras, ultraviolet cameras, night vision cameras, or even ambient light sensors. Ambient light sensors may optionally

19

be coupled to the main processor **505** and/or the display screen **520** to control brightness of the display screen **520** and/or of a backlight of the display screen **520** in order to acclimatize to the ambient light level around the payment reader terminal device **125** and prevent the display screen **520** from appearing too bright/harsh or too dim/unclear. The additional or alternative elements may include biometric sensors, such as fingerprint scanners, handprint scanners, voice authentication, facial recognition or iris scanners (using any of the camera types discussed above), blood testers, DNA testers, or some combination thereof. The processor(s) **505/110** may compare received biometric data from the buyer via the biometric sensors against databases stored within the memory **535** or accessible to the payment reader terminal device **125** via a network connection via the Internet or a private network. The additional or alternative elements may include one or more accelerometer(s) and/or one or more gyroscope(s), allowing the processor(s) **505/110** of the payment reader terminal device **125** to react to being tilted, moved, or shaken, optionally as indications of tampering via the tamper protection circuitry **530** if the payment reader terminal device **125** is intended to be stationary. The additional or alternative elements may include one or more indicator lights, which may be light-emitting diodes (LED) and may emit light, optionally at varying brightnesses and/or colors, to indicate a successful transaction, to indicate a failed transaction, to indicate that a card or other payment instrument should be presented/inserted/swiped, to indicate that a card or other payment instrument should be removed, to indicate that a user input such as a PIN code must be input via the touch-sensitive display screen **210**, to indicate that an input via the touch-sensitive display screen **210** has been received, to indicate successful connection via any of the antennae of the POS system **500**, to indicate failure to connect via any of the antennae of the POS system **500**, to indicate a signal strength associated with any of the antennae of the POS system **500**, or some combination thereof. The additional or alternative elements may include any components illustrated in or discussed with respect to FIG. 7.

It should be noted that the components of the payment reader terminal device **125** of FIG. 5 are just exemplary. In some cases, any subset of these may be missing from the payment reader terminal device **125**. For example, a payment reader terminal device **125** may be a “headless” device that has no display screen **520** or touch-sensitive surface interface **525**, and may instead include one or more light emitting diodes (LEDs) as indicators of when to insert a payment card or approach with a NFC-enabled payment instrument and/or one or more speakers and/or one or more physical buttons, or may include none of these things. The payment reader terminal device **125** may have any number of payment instrument readers **515**—it may have one or more magnetic stripe readers, one or more IC chip readers, one or more NFC readers, or any combination thereof.

FIG. 6 illustrates a system architecture including a web-based point of sale (POS) device and mobile device that collaborate to perform a cardless transaction. The system architecture **600** of FIG. 6 includes, for example, the checkout service **110**, a web-based POS system **620**, and a mobile device **625**.

In some cases, the merchant POS device **120** may be a web-based POS system **620** hosted on a website rather than a device physically located at the merchant location **105**. In this case, the merchant POS device **120** as discussed otherwise herein may be a web server that serves the website, a browser device that browses the website, or some combination thereof. That is, the web server or the browser device

20

may fulfill the role of the merchant POS device **120** in the contexts discussed herein, for example in the context of the architecture **100**, the operations **200**, the operations **300**, the operations **400**, or some combination thereof.

In this case, the web-based POS **620** may receive an input from the merchant or customer interacting with the website to select one or more purchases of items and/or services by the customer from the merchant. The web-based POS **620**—that is, the web server or the browser device—may send a request (signified by arrow **630**) to the checkout service **110** to process the payment amount of the transaction, which may be received by the checkout service **110**, much like steps **205** and **210** of the operations **200** of FIG. 2.

The checkout service **110** may, as signified by arrow **635**, output a quick response (QR) code **605** in response to receipt of the request at arrow **630**. As signified by arrow **640**, the QR code **605** may be sent to the web server and/or browser device so that it appears in the web-based POS **620** and is displayed by the device browsing the web-based POS **620**, for example via an iFrame in the web-based POS **620** or PHP insert into web-based POS **620**. A customer can use his or her mobile device **625** to scan the QR code **605** with a camera of the mobile device **625** as signified by the arrow **645**. The QR code **605**, as generated by the checkout service **110**, may lead to a webpage associated with the checkout service **110** through which the customer may perform a cardless checkout, for example by entering payment instrument information such as credit or debit card numbers, by entering a login associated with the checkout service to access previously-entered or previously-read payment instrument information. The customer can then approve of the payment and directly notify the checkout service (as signified by arrow **650**) to proceed with transfer of funds from an account associated with the customer to an account associated with the merchant. The checkout service **110** may then perform the transfer or request that one or more financial entity server(s) **135** perform the transfer as in steps **245**, **250**, and **252** of the operations **200** of FIG. 2. Like steps **255**, **260**, **265**, and **270** of the operations **200** of FIG. 2, the checkout service **110** may receive and send confirmations that the transfer has completed successfully and that the transaction has been processed successfully.

The QR code **605** may alternately be a bar code or a text link that may be entered into the mobile device **625** manually.

FIG. 7 illustrates an exemplary computing system **700** that may be used to implement some aspects of the technology. For example, any of the computing devices, computing systems, network devices, network systems, servers, and/or arrangements of circuitry described herein may include at least one computing system **700**, or may include at least one component of the computer system **700** identified in FIG. 7. The computing system **700** of FIG. 7 includes one or more processors **710** and memory **720**. Each of the processor(s) **710** may refer to one or more processors, controllers, microcontrollers, central processing units (CPUs), graphics processing units (GPUs), arithmetic logic units (ALUs), accelerated processing units (APUs), digital signal processors (DSPs), application specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), or combinations thereof. Each of the processor(s) **710** may include one or more cores, either integrated onto a single chip or spread across multiple chips connected or coupled together. Memory **720** stores, in part, instructions and data for execution by processor **710**. Memory **720** can store the executable code when in operation. The system **700** of FIG. 7 further includes a mass storage device **730**, portable

storage medium drive(s) **740**, output devices **750**, user input devices **760**, a graphics display **770**, and peripheral devices **780**.

The components shown in FIG. 7 are depicted as being connected via a single bus **790**. However, the components may be connected through one or more data transport means. For example, processor unit **710** and memory **720** may be connected via a local microprocessor bus, and the mass storage device **730**, peripheral device(s) **780**, portable storage device **740**, and display system **770** may be connected via one or more input/output (I/O) buses.

Mass storage device **730**, which may be implemented with a magnetic disk drive or an optical disk drive, is a non-volatile storage device for storing data and instructions for use by processor unit **710**. Mass storage device **730** can store the system software for implementing some aspects of the subject technology for purposes of loading that software into memory **720**.

Portable storage device **740** operates in conjunction with a portable non-volatile storage medium, such as a floppy disk, compact disk or Digital video disc, to input and output data and code to and from the computer system **700** of FIG. 7. The system software for implementing aspects of the subject technology may be stored on such a portable medium and input to the computer system **700** via the portable storage device **740**.

The memory **720**, mass storage device **730**, or portable storage device **740** may in some cases store sensitive information, such as transaction information, health information, or cryptographic keys, and may in some cases encrypt or decrypt such information with the aid of the processor **710**. The memory **720**, mass storage device **730**, or portable storage device **740** may in some cases store, at least in part, instructions, executable code, or other data for execution or processing by the processor **710**.

Output devices **750** may include, for example, communication circuitry for outputting data through wired or wireless means, display circuitry for displaying data via a display screen, audio circuitry for outputting audio via headphones or a speaker, printer circuitry for printing data via a printer, or some combination thereof. The display screen may be any type of display discussed with respect to the display system **770**. The printer may be inkjet, laserjet, thermal, or some combination thereof. In some cases, the output device circuitry **750** may allow for transmission of data over an audio jack/plug, a microphone jack/plug, a universal serial bus (USB) port/plug, an Apple® Lightning® port/plug, an Ethernet port/plug, a fiber optic port/plug, a proprietary wired port/plug, a BLUETOOTH® wireless signal transfer, a BLUETOOTH® low energy (BLE) wireless signal transfer, an IBEACON® wireless signal transfer, a radio-frequency identification (RFID) wireless signal transfer, near-field communications (NFC) wireless signal transfer, dedicated short range communication (DSRC) wireless signal transfer, 802.11 Wi-Fi wireless signal transfer, wireless local area network (WLAN) signal transfer, Visible Light Communication (VLC), Worldwide Interoperability for Microwave Access (WiMAX), Infrared (IR) communication wireless signal transfer, Public Switched Telephone Network (PSTN) signal transfer, Integrated Services Digital Network (ISDN) signal transfer, 3G/4G/5G/LTE cellular data network wireless signal transfer, ad-hoc network signal transfer, radio wave signal transfer, microwave signal transfer, infrared signal transfer, visible light signal transfer, ultraviolet light signal transfer, wireless signal transfer along the electromagnetic spectrum, or some combination thereof. Output devices **750** may include any ports, plugs, antennae, wired

or wireless transmitters, wired or wireless transceivers, or any other components necessary for or usable to implement the communication types listed above, such as cellular Subscriber Identity Module (SIM) cards.

Input devices **760** may include circuitry providing a portion of a user interface. Input devices **760** may include an alpha-numeric keypad, such as a keyboard, for inputting alpha-numeric and other information, or a pointing device, such as a mouse, a trackball, stylus, or cursor direction keys. Input devices **760** may include touch-sensitive surfaces as well, either integrated with a display as in a touchscreen, or separate from a display as in a trackpad. Touch-sensitive surfaces may in some cases detect localized variable pressure or force detection. In some cases, the input device circuitry may allow for receipt of data over an audio jack, a microphone jack, a universal serial bus (USB) port/plug, an Apple® Lightning® port/plug, an Ethernet port/plug, a fiber optic port/plug, a proprietary wired port/plug, a wired local area network (LAN) port/plug, a BLUETOOTH® wireless signal transfer, a BLUETOOTH® low energy (BLE) wireless signal transfer, an IBEACON® wireless signal transfer, a radio-frequency identification (RFID) wireless signal transfer, near-field communications (NFC) wireless signal transfer, dedicated short range communication (DSRC) wireless signal transfer, 802.11 Wi-Fi wireless signal transfer, wireless local area network (WLAN) signal transfer, Visible Light Communication (VLC), Worldwide Interoperability for Microwave Access (WiMAX), Infrared (IR) communication wireless signal transfer, Public Switched Telephone Network (PSTN) signal transfer, Integrated Services Digital Network (ISDN) signal transfer, 3G/4G/5G/LTE cellular data network wireless signal transfer, personal area network (PAN) signal transfer, wide area network (WAN) signal transfer, ad-hoc network signal transfer, radio wave signal transfer, microwave signal transfer, infrared signal transfer, visible light signal transfer, ultraviolet light signal transfer, wireless signal transfer along the electromagnetic spectrum, or some combination thereof. Input devices **760** may include any ports, plugs, antennae, wired or wireless receivers, wired or wireless transceivers, or any other components necessary for or usable to implement the communication types listed above, such as cellular SIM cards.

Input devices **760** may include receivers or transceivers used for positioning of the computing system **700** as well. These may include any of the wired or wireless signal receivers or transceivers. For example, a location of the computing system **700** can be determined based on signal strength of signals as received at the computing system **700** from three cellular network towers, a process known as cellular triangulation. Fewer than three cellular network towers can also be used—even one can be used—though the location determined from such data will be less precise (e.g., somewhere within a particular circle for one tower, somewhere along a line or within a relatively small area for two towers) than via triangulation. More than three cellular network towers can also be used, further enhancing the location's accuracy. Similar positioning operations can be performed using proximity beacons, which might use short-range wireless signals such as BLUETOOTH® wireless signals, BLUETOOTH® low energy (BLE) wireless signals, IBEACON® wireless signals, personal area network (PAN) signals, microwave signals, radio wave signals, or other signals discussed above. Similar positioning operations can be performed using wired local area networks (LAN) or wireless local area networks (WLAN) where locations are known of one or more network devices in

23

communication with the computing system **700** such as a router, modem, switch, hub, bridge, gateway, or repeater. These may also include Global Navigation Satellite System (GNSS) receivers or transceivers that are used to determine a location of the computing system **700** based on receipt of one or more signals from one or more satellites associated with one or more GNSS systems. GNSS systems include, but are not limited to, the US-based Global Positioning System (GPS), the Russia-based Global Navigation Satellite System (GLONASS), the China-based BeiDou Navigation Satellite System (BDS), and the Europe-based Galileo GNSS. Input devices **760** may include receivers or transceivers corresponding to one or more of these GNSS systems.

Display system **770** may include a liquid crystal display (LCD), a plasma display, an organic light-emitting diode (OLED) display, an electronic ink or “e-paper” display, a projector-based display, a holographic display, or another suitable display device. Display system **770** receives textual and graphical information, and processes the information for output to the display device. The display system **770** may include multiple-touch touchscreen input capabilities, such as capacitive touch detection, resistive touch detection, surface acoustic wave touch detection, or infrared touch detection. Such touchscreen input capabilities may or may not allow for variable pressure or force detection.

Peripherals **780** may include any type of computer support device to add additional functionality to the computer system. For example, peripheral device(s) **780** may include one or more additional output devices of any of the types discussed with respect to output device **750**, one or more additional input devices of any of the types discussed with respect to input device **760**, one or more additional display systems of any of the types discussed with respect to display system **770**, one or more memories or mass storage devices or portable storage devices of any of the types discussed with respect to memory **720** or mass storage **730** or portable storage **740**, a modem, a router, an antenna, a wired or wireless transceiver, a printer, a bar code scanner, a quick-response (“QR”) code scanner, a magnetic stripe card reader, an integrated circuit chip (ICC) card reader such as a smart-card reader or a EUROPAY®-MASTERCARD®-VISA® (EMV) chip card reader, a near field communication (NFC) reader, a document/image scanner, a visible light camera, a thermal/infrared camera, an ultraviolet-sensitive camera, a night vision camera, a light sensor, a phototransistor, a photoresistor, a thermometer, a thermistor, a battery, a power source, a proximity sensor, a laser rangefinder, a sonar transceiver, a radar transceiver, a lidar transceiver, a network device, a motor, an actuator, a pump, a conveyor belt, a robotic arm, a rotor, a drill, a chemical assay device, or some combination thereof.

The components contained in the computer system **700** of FIG. 7 can include those typically found in computer systems that may be suitable for use with some aspects of the subject technology and represent a broad category of such computer components that are well known in the art. That said, the computer system **700** of FIG. 7 can be customized and specialized for the purposes discussed herein and to carry out the various operations discussed herein, with specialized hardware components, specialized arrangements of hardware components, and/or specialized software. Thus, the computer system **700** of FIG. 7 can be a personal computer, a hand held computing device, a telephone (“smartphone” or otherwise), a mobile computing device, a workstation, a server (on a server rack or otherwise), a minicomputer, a mainframe computer, a tablet computing

24

device, a wearable device (such as a watch, a ring, a pair of glasses, or another type of jewelry or clothing or accessory), a video game console (portable or otherwise), an e-book reader, a media player device (portable or otherwise), a vehicle-based computer, another type of computing device, or some combination thereof. The computer system **700** may in some cases be a virtual computer system executed by another computer system. The computer can also include different bus configurations, networked platforms, multi-processor platforms, etc. Various operating systems can be used including Unix®, Linux®, FreeBSD®, FreeNAS®, pfSense®, Windows®, Apple® Macintosh OS® (“MacOS®”), Palm OS®, Google® Android®, Google® Chrome OS®, Chromium® OS®, OPENSTEP®, XNU®, Darwin®, Apple® iOS®, Apple® tvOS®, Apple® watchOS®, Apple® audioOS®, Amazon® Fire OS®, Amazon® Kindle OS®, variants of any of these, other suitable operating systems, or combinations thereof. The computer system **700** may also use a Basic Input/Output System (BIOS) or Unified Extensible Firmware Interface (UEFI) as a layer upon which the operating system(s) are run.

In some cases, the computer system **700** may be part of a multi-computer system that uses multiple computer systems **700**, each for one or more specific tasks or purposes. For example, the multi-computer system may include multiple computer systems **700** communicatively coupled together via at least one of a personal area network (PAN), a local area network (LAN), a wireless local area network (WLAN), a municipal area network (MAN), a wide area network (WAN), or some combination thereof. The multi-computer system may further include multiple computer systems **700** from different networks communicatively coupled together via the internet (also known as a “distributed” system).

Some aspects of the subject technology may be implemented in an application that may be operable using a variety of devices. Non-transitory computer-readable storage media refer to any medium or media that participate in providing instructions to a central processing unit (CPU) for execution and that may be used in the memory **720**, the mass storage **730**, the portable storage **740**, or some combination thereof. Such media can take many forms, including, but not limited to, non-volatile and volatile media such as optical or magnetic disks and dynamic memory, respectively. Some forms of non-transitory computer-readable media include, for example, a floppy disk, a flexible disk, a hard disk, magnetic tape, a magnetic strip/stripe, any other magnetic storage medium, flash memory, memristor memory, any other solid-state memory, a compact disc read only memory (CD-ROM) optical disc, a rewritable compact disc (CD) optical disc, digital video disk (DVD) optical disc, a blu-ray disc (BDD) optical disc, a holographic optical disc, another optical medium, a secure digital (SD) card, a micro secure digital (microSD) card, a Memory Stick® card, a smartcard chip, a EMV chip, a subscriber identity module (SIM) card, a mini/micro/nano/pico SIM card, another integrated circuit (IC) chip/card, random access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash EPROM (FLASH/EPROM), cache memory (L1/L2/L3/L4/L5/L7), resistive random-access memory (RRAM/ReRAM), phase change memory (PCM), spin transfer torque RAM (STT-RAM), another memory chip or cartridge, or a combination thereof.

25

Various forms of transmission media may be involved in carrying one or more sequences of one or more instructions to a processor 710 for execution. A bus 790 carries the data to system RAM or another memory 720, from which a processor 710 retrieves and executes the instructions. The instructions received by system RAM or another memory 720 can optionally be stored on a fixed disk (mass storage device 730/portable storage 740) either before or after execution by processor 710. Various forms of storage may likewise be implemented as well as the necessary network interfaces and network topologies to implement the same.

While various flow diagrams provided and described above may show a particular order of operations performed by some embodiments of the subject technology, it should be understood that such order is exemplary. Alternative embodiments may perform the operations in a different order, combine certain operations, overlap certain operations, or some combination thereof. It should be understood that unless disclosed otherwise, any process illustrated in any flow diagram herein or otherwise illustrated or described herein may be performed by a machine, mechanism, and/or computing system 700 discussed herein, and may be performed automatically (e.g., in response to one or more triggers/conditions described herein), autonomously, semi-autonomously (e.g., based on received instructions), or a combination thereof. Furthermore, any action described herein as occurring in response to one or more particular triggers/conditions should be understood to optionally occur automatically response to the one or more particular triggers/conditions.

The foregoing detailed description of the technology has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the technology to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. The described embodiments were chosen in order to best explain the principles of the technology, its practical application, and to enable others skilled in the art to utilize the technology in various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the scope of the technology be defined by the claim.

The invention claimed is:

1. A method of identifying devices for pairing, the method comprising:

receiving a request from a first merchant device of a merchant;

identifying that a first identifier associated with the first merchant device is included in a database associated with a checkout service;

identifying that the first merchant device is located in an area associated with the merchant;

identifying a second merchant device that is within the area and that is associated with a second identifier that is included in the database associated with the checkout service, wherein the second merchant device includes a payment instrument reader;

sending a response to the first merchant device of the merchant, the response identifying that the second merchant device is available to be paired to the first merchant device through the checkout service based on the first merchant device and the second merchant device being co-located within the area and based on the first identifier and the second identifier both being included in the database associated with the checkout service; and

facilitating a transaction using the first merchant device and the second merchant device while the first mer-

26

chant device and the second merchant device are paired together through the checkout service, wherein facilitating the transaction includes initiating processing of the transaction using payment information read from a payment instrument using the payment instrument reader of the second merchant device.

2. The method of claim 1, further comprising:

identifying a third merchant device that is within the area and that is associated with a third identifier that is included in the database associated with the checkout service, wherein the response also identifies the third merchant device as available to be paired to the first merchant device through the checkout service based on the first merchant device and the third merchant device being co-located within the area and based on the first identifier and the third identifier both being included in the database associated with the checkout service.

3. The method of claim 1, wherein the area is associated with a store of the merchant.

4. The method of claim 1, further comprising:

receiving a confirmation that the second merchant device is to be paired to the first merchant device before the facilitating of the transaction using the first merchant device and the second merchant device.

5. A method of identifying devices for pairing, the method comprising:

receiving a request from a first merchant device of a merchant;

identifying that the first merchant device is located in an area associated with the merchant;

identifying a second merchant device that is within the area and that is associated with a second identifier, wherein the second merchant device includes a payment instrument reader;

sending a response to the first merchant device of the merchant, the response identifying that the second merchant device is available to be paired to the first merchant device through a checkout service based on the first merchant device and the second merchant device being co-located within the area; and

facilitating a transaction using the first merchant device and the second merchant device while the first merchant device and the second merchant device are paired together through the checkout service, wherein facilitating the transaction includes initiating processing of the transaction using payment information read from a payment instrument using the payment instrument reader of the second merchant device.

6. The method of claim 5, further comprising:

identifying that a first identifier associated with the first merchant device is included in a database associated with the checkout service, wherein the second identifier is included in the database associated with the checkout service, and wherein the response identifies that the second merchant device is available to be paired to the first merchant device through the checkout service also based on the first identifier and the second identifier both being included in the database associated with the checkout service.

7. The method of claim 6, further comprising:

identifying a third merchant device that is within the area and that is associated with a third identifier that is included in the database associated with the checkout service, wherein the response also identifies the third merchant device as available to be paired to the first merchant device through the checkout service based on the first merchant device and the third merchant device

27

being co-located within the area and based on the first identifier and the third identifier both being included in the database associated with the checkout service.

8. The method of claim 7, further comprising:

receiving a selection of the second merchant device from a set that includes the second merchant device and the third merchant device, wherein the selection indicates that the second merchant device is to be paired to the first merchant device before the facilitating of the transaction using the first merchant device and the second merchant device.

9. The method of claim 5, wherein the area is associated with a store of the merchant.

10. The method of claim 5, further comprising:

receiving a confirmation that the second merchant device is to be paired to the first merchant device before the facilitating of the transaction using the first merchant device and the second merchant device.

11. The method of claim 10, wherein the first merchant device identifies at least one of a product or service associated with the transaction.

12. The method of claim 5, wherein the first merchant device includes a second payment instrument reader, wherein facilitating the transaction includes initiating the processing of the transaction using secondary payment information read from a second payment instrument using the second payment instrument reader of the first merchant device.

13. The method of claim 12, wherein the second merchant device identifies at least one of a product or service associated with the transaction.

14. The method of claim 5, wherein the response identifies a plurality of merchant devices that are available to be paired to the first merchant device through the checkout service based on the first merchant device and the plurality of merchant devices being co-located within the area, wherein the second merchant device is one of the plurality of merchant devices.

15. The method of claim 14, further comprising:

receiving a selection of the second merchant device from the plurality of merchant devices, wherein the selection indicates that the second merchant device is to be paired to the first merchant device before the facilitating of the transaction using the first merchant device and the second merchant device.

16. A system for identifying devices for pairing, the system comprising:

one or more memory units that store instructions; and one or more processors that execute the instructions, wherein execution of the instructions by the one or more processors causes the one or more processors to: receive a request from a first merchant device of a merchant;

28

identify that the first merchant device is located in an area associated with the merchant;

identify a second merchant device that is within the area and that is associated with a second identifier, wherein the second merchant device includes a payment instrument reader;

send a response to the first merchant device of the merchant, the response identifying that the second merchant device is available to be paired to the first merchant device through a checkout service based on the first merchant device and the second merchant device being co-located within the area; and

facilitate a transaction using the first merchant device and the second merchant device while the first merchant device and the second merchant device are paired together through the checkout service, wherein facilitating the transaction includes initiating processing of the transaction using payment information read from a payment instrument using the payment instrument reader of the second merchant device.

17. The system of claim 16, wherein the execution of the instructions by the one or more processors causes the one or more processors to:

identify that a first identifier associated with the first merchant device is included in a database associated with the checkout service, wherein the second identifier is included in the database associated with the checkout service, and wherein the response identifies that the second merchant device is available to be paired to the first merchant device through the checkout service also based on the first identifier and the second identifier both being included in the database associated with the checkout service.

18. The system of claim 16, wherein the area is associated with a store of the merchant.

19. The system of claim 16, wherein the execution of the instructions by the one or more processors causes the one or more processors to:

receive a confirmation that the second merchant device is to be paired to the first merchant device before the facilitating of the transaction using the first merchant device and the second merchant device.

20. The system of claim 16, wherein the response identifies a plurality of merchant devices that are available to be paired to the first merchant device through the checkout service based on the first merchant device and the plurality of merchant devices being co-located within the area, wherein the second merchant device is one of the plurality of merchant devices.

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